Unit of Observation

Each row of the Spotify reviews dataset represents a single user-submitted review of the Spotify mobile application. A review consists of the user's written text feedback, a numeric star rating, and associated metadata such as the review date or app version if present. This dataset is used to analyze sentiment expressed in text reviews and compare it to the corresponding star rating.

Dataset Summary

- Total observations: 61,594 reviews
- Data type: Text and numeric ratings
- Source: Kaggle Spotify Reviews dataset
- Variables include:
  - Original review text
  - Cleaned review text (emoji removed)
  - User rating (1–5)
  - VADER compound sentiment score (−1 to +1)
  - Predicted ratings from VADER and Hugging Face
  - Error metrics for model evaluation

Variable Descriptions

Below is a complete list of all variables analyzed in this project. For each variable, we provide the description, data type, possible values or range, and summary statistics where applicable.

Review

Description:
The written text of the user's review of the Spotify app as submitted in the app store. This variable contains the original, unprocessed natural language feedback.

Data Type:
Text (string)

Example Values:
"Excellent great music app 👍."
"App crashes every time I open it"

Summary Statistics:
Total number of reviews: [61,694]
Number of non-missing reviews: [61,594]

Notes:
This is the raw text variable used prior to preprocessing. It serves as the basis for emoji cleaning and sentiment scoring.

Rating

Description:
The numeric star rating assigned by the user to the Spotify app at the time of review submission.

Data Type:
Integer (1–5)

Example Values:
1, 2, 3, 4, 5

Summary Statistics:
Mean rating: Approximately 4.0
Distribution skewed toward: Higher ratings, 4-5 star reviews

Notes:
This variable represents the ground-truth outcome used to evaluate model prediction accuracy.

Review_compound

Description:
The VADER compound sentiment score representing overall sentiment of the review. This value combines positive, negative, and neutral components into a single normalized score.

Data Type:
Continuous numeric (−1 to +1)

Example Values:
−0.263
0.579

Summary Statistics:
Mean compound score: 0.280
Range: −1 to +1

Notes:
Scores near −1 indicate strong negative sentiment, scores near +1 indicate strong positive sentiment. Values near 0 indicate neutral sentiment.

Review_cleaned

Description:
The review text after emoji characters were removed during preprocessing to assess whether emojis affected sentiment model accuracy.

Data Type:
Text (string)

Example Values:
"Excellent great music app."

Summary Statistics:
Number of cleaned reviews: 61,594

Notes:
Emoji removal was performed to standardize input text for sentiment models and reduce noise introduced by symbolic characters.

vader_predicted_rating

Description:
The predicted star rating derived from the VADER compound sentiment score using a linear transformation from the −1 to +1 scale to the 1–5 rating scale.

Data Type:
Integer (1–5)

Example Values:
1, 2, 3, 4, 5

Summary Statistics:
Exact match accuracy: 31.66%
Average error: 1.09

Notes:
This variable allows direct comparison between VADER-derived sentiment and the user's actual rating.

hf_predicted_rating

Description:
The predicted star rating generated by the Hugging Face BERT transformer model based on the review text.

Data Type:
Integer (1–5)

Example Values:
1, 2, 3, 4, 5

Summary Statistics:
Exact match accuracy: 58.63%
Average error: 0.62

Notes:
This model uses contextual embeddings and generally demonstrates improved predictive performance compared to VADER.

Exact_match %

Description:
The percentage of reviews for which the predicted rating exactly matched the user's actual star rating.

Data Type:
Percentage (0–100%)

Example Values:
31.66% (VADER)
58.63% (Hugging Face)

Notes:
This metric evaluates strict prediction accuracy.

vader_error

Description:
The absolute difference between the actual user rating and the rating predicted by VADER.

Data Type:
Integer (0–4)

Example Values:
0, 1, 2, 3

Summary Statistics:
Mean error: 1.09

Notes:
A value of 0 indicates an exact match, higher values indicate greater deviation.

hf_error

Description:
The absolute difference between the actual user rating and the rating predicted by the Hugging Face model.

Data Type:
Integer (0–4)

Example Values:
0, 1, 2, 3

Summary Statistics:
Mean error: 0.62

Notes:
Lower average error indicates stronger predictive alignment with user ratings.

Off-by-One %

Description:
The percentage of predictions that were within one star (±1) of the actual rating.

Data Type:
Percentage (0–100%)

Example Values:
71.85% (VADER)
86.55% (Hugging Face)

Notes:
This metric accounts for near-miss predictions and provides a more flexible measure of performance than exact match accuracy.

Avg Error

Description:
The average absolute difference between predicted ratings and actual ratings.

Data Type:
Continuous numeric

Example Values:
1.09 (VADER)
0.62 (Hugging Face)

Notes:
This metric summarizes overall prediction deviation across all reviews.

Exploratory Figures and Tables

The following figures were created during data establishment.

Figure A: Sentiment Distribution Across Review Length Categories

Description:
This figure shows the distribution of VADER compound sentiment scores across five review length categories for all 61,594 reviews.

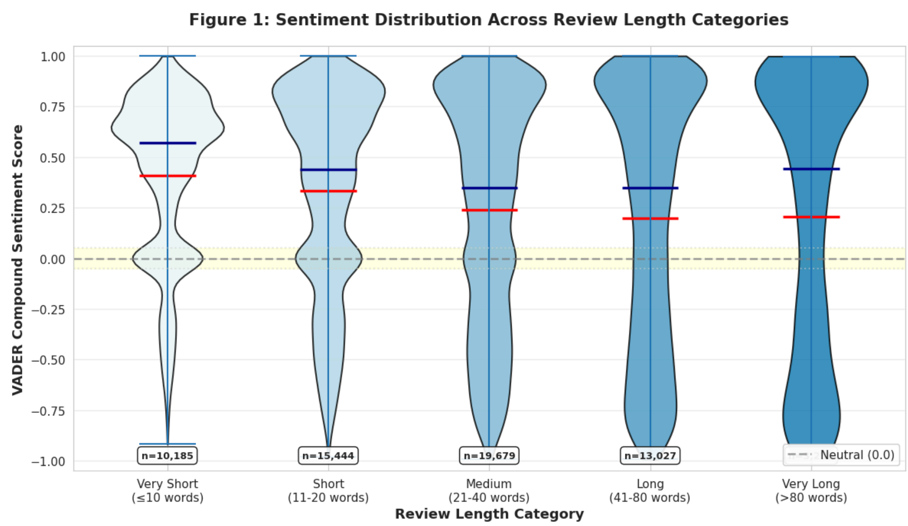


Figure B: Distribution of VADER Compound Sentiment Scores

Description:
This histogram displays the overall distribution of VADER compound sentiment scores across 61,594 reviews.

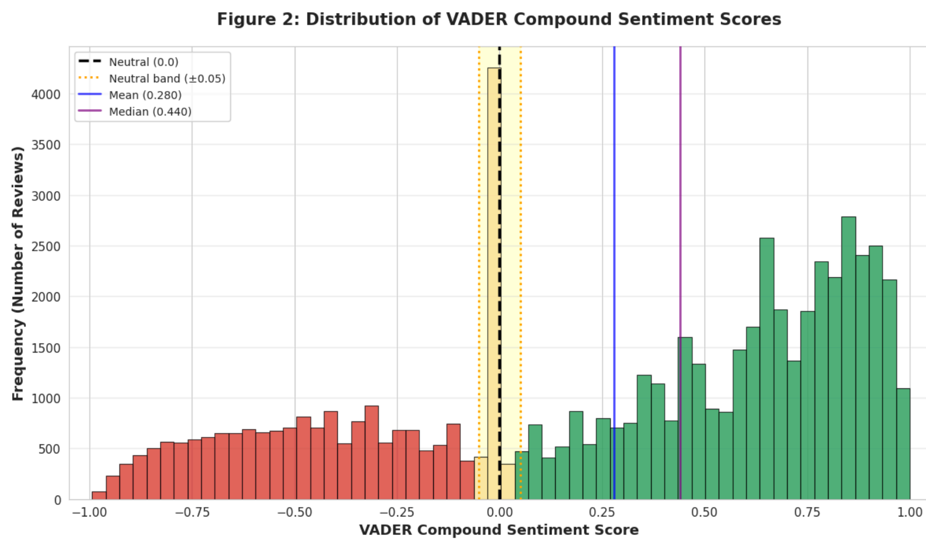
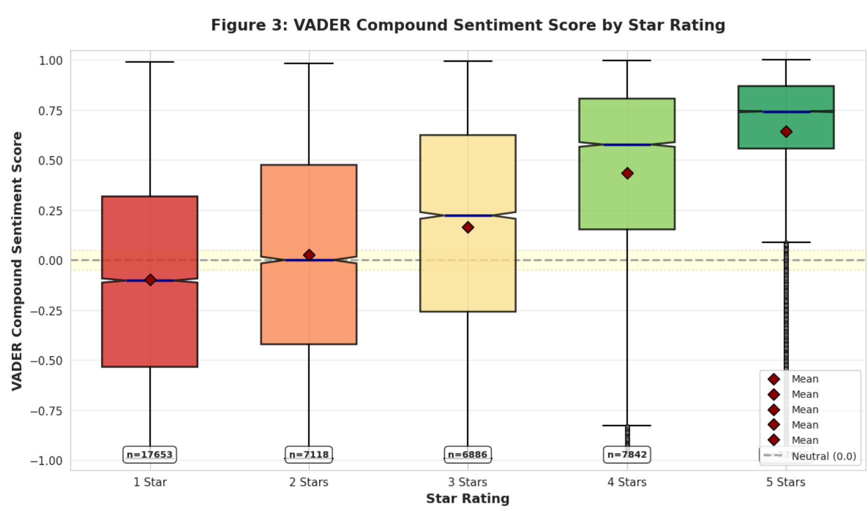


Figure C: Compound Sentiment by Star Rating

Description:

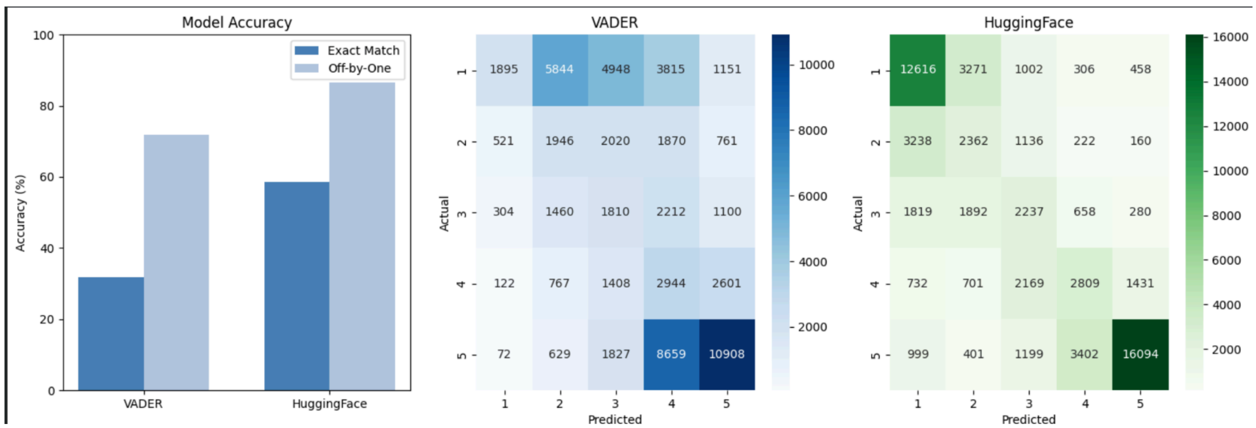
This boxplot shows compound sentiment scores grouped by star rating (1–5 stars).



Model Output and Evaluation

The following figures were produced to evaluate model performance.

Confusion Matrices (With Emojis)



Confusion matrices compare actual ratings to predicted ratings for both VADER and Hugging Face. Hugging Face predictions show stronger diagonal concentration, indicating improved exact classification accuracy.

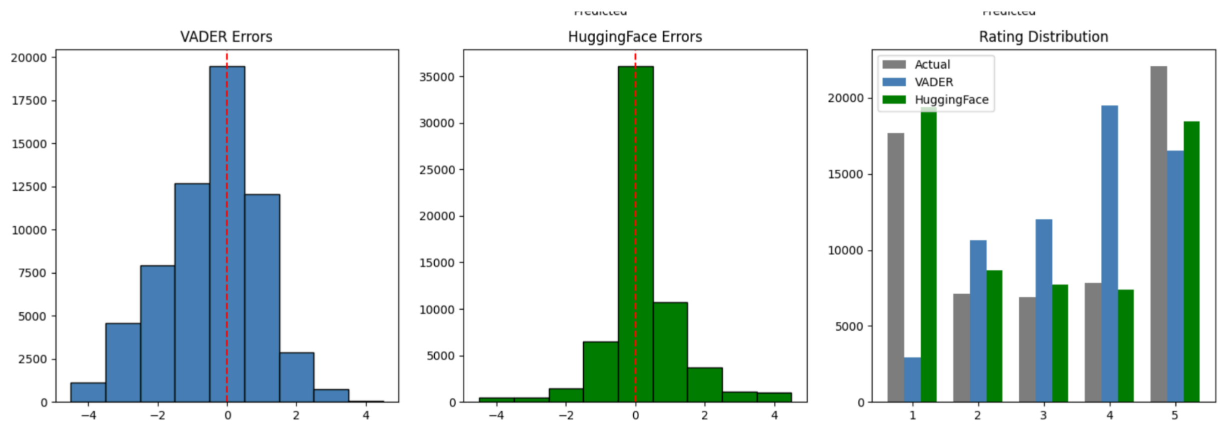
Model Performance Metrics (With Emojis)

	Model	Exact Match %	Off-by-One %	Avg Error
0	VADER	31.663798	71.805695	1.091502
1	HuggingFace	58.638828	86.558756	0.621294

These metrics quantify model uncertainty and predictive accuracy.

- Exact Match % measures strict prediction accuracy.
- Off-by-One % allows ±1 rating flexibility.
- Average Error measures mean absolute deviation.

Error Distributions (With Emojis)



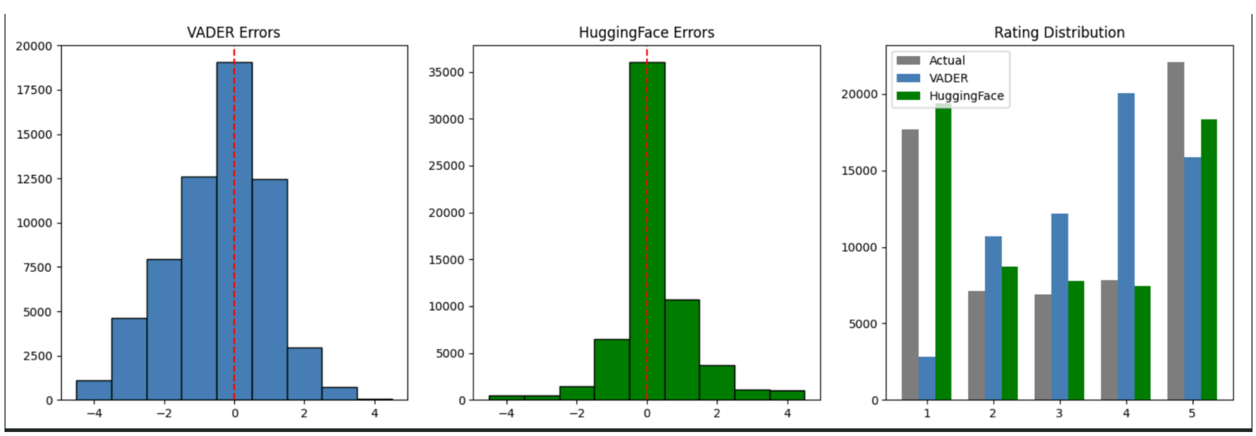
Histograms display the distribution of prediction errors for both models. Hugging Face errors cluster more tightly around zero, indicating improved predictive precision.

Model Performance Metrics (Without Emojis)

	Model	Exact Match %	Off-by-One %	Avg Error
0	VADER	30.972173	71.677436	1.098954
1	HuggingFace	58.572264	86.592850	0.621359

This suggests emoji removal had limited impact on overall predictive performance.

Error Distributions (Without Emojis)



The error distribution histograms show that Hugging Face prediction errors remain tightly concentrated around zero, while VADER errors are more dispersed, indicating that removing emojis does not substantially change overall model performance or relative accuracy.

Notes on Data Processing

- Reviews with missing text or missing star ratings were excluded from analysis.
- Ratings outside 1–5 were dropped.
- All variables were generated using scripts in the SCRIPTS folder (project1.ipynb and official_code.ipynb).
- Engineered variables (review_length_words, review_length_chars, and sentiment scores) were computed in preprocessing and sentiment scoring steps.